Group ID	6
Student ID	230042104, 230042110, 230042118
Title	Plastic Box Solver
Brief Description	A browser-based Rubik's Cube scanning and solving assistant. Users scan or manually input their physical cube's state, the system validates and renders it as an interactive 3D model, then computes a solution (optimal via the Kociemba two-phase algorithm, or beginner's-method for guided learning). Users can solve interactively with a live timer, follow guided step-by-step instructions, or watch an automatic solve animation, with results (time, move count, efficiency) tracked, saved to history, and ranked on a global leaderboard.

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1	FR-01	The system shall allow a new user to register for an account using email and password.	US-01	New User Registration	first-time visitor to the platform	to create an account using my email address and a password	I can save my scan history, solve times, and leaderboard rank across sessions	As a first-time visitor to the platform, I want to create an account using my email address and a password, so that I can save my scan history, solve times, and leaderboard rank across sessions.	• Given a visitor enters a valid, unused email and a password of at least 8 characters, when they submit the registration form, then an account is created and the user is redirected to the Main Menu. • Given a visitor enters an email that already exists in the system, when they submit the form, then the system displays 'Email already registered' and does not create a duplicate account. • Given a visitor enters a password shorter than 8 characters, when they submit the form, then the system blocks submission and shows an inline validation error. • Given a registration succeeds, when the password is persisted, then it is stored only in hashed form, never in plaintext.	Medium	To Do				
2	FR-02	The system shall allow a registered user to log in using their credentials.	US-02	Log In to Existing Account	registered user returning to the app	to log in with my email and password	I can resume practicing without losing my saved history or leaderboard standing	As a registered user returning to the app, I want to log in with my email and password, so that I can resume practicing without losing my saved history or leaderboard standing.	• Given a registered user enters correct credentials, when they submit the login form, then they are authenticated and taken to the Main Menu. • Given a user is successfully authenticated, when the session is created, then it persists across page reloads until logout or expiry. • Given an unauthenticated user tries to access a protected page directly by URL, when the page loads, then they are redirected to the login screen.	Medium	To Do				
3	FR-03	The system shall allow a logged-in user to view and update their personal profile.	US-03	Manage Personal Profile	logged-in user	to view and edit my profile details such as display name and avatar	my identity is shown correctly on my solve history and the leaderboard	As a logged-in user, I want to view and edit my profile details such as display name and avatar, so that my identity is shown correctly on my solve history and the leaderboard.	• Given a user opens the profile page, when it loads, then their current display name and email are shown. • Given a user changes their display name to a valid, unique value, when they save, then the new name is reflected immediately on the leaderboard. • Given a user attempts to set a display name already taken by another user, when they save, then the system rejects the change with an explanatory error. • Given a user updates their profile, when the update completes, then a success confirmation is shown.	Low	To Do				
4	FR-04	The system shall accept a scrambled cube state captured via the device camera.	US-04	Scan Physical Cube with Camera	user who owns a physical Rubik's Cube	to scan each of the six faces using my phone or webcam camera	I don't have to manually enter every sticker color by hand	As a user who owns a physical Rubik's Cube, I want to scan each of the six faces using my phone or webcam camera, so that I don't have to manually enter every sticker color by hand.	• Given the user grants camera permission, when they align a cube face inside the on-screen guide frame, then the system captures the image for that face. • Given all six faces have been captured, when the last face is submitted, then the backend receives all six images for color processing. • Given camera permission is denied, when the user opens the scan screen, then the system shows an explanatory message and offers the manual input option. • Given a captured image is blurry or too dark to process, when it is submitted, then the system prompts the user to retake that specific face.	High	In Progress				
5	FR-05	The system shall provide a manual click-input interface for setting sticker colors when camera scanning is unreliable.	US-05	Manually Enter Cube Colors	user in poor lighting conditions	to set each of the 54 sticker colors by clicking on a flat 2D cube net	I can still submit an accurate cube state when camera scanning fails	As a user in poor lighting conditions, I want to set each of the 54 sticker colors by clicking on a flat 2D cube net, so that I can still submit an accurate cube state when camera scanning fails.	• Given the manual input screen is open, when the user clicks a sticker cell, then a color palette appears for selecting that sticker's color. • Given the user has not yet assigned a color to every one of the 54 stickers, when they attempt to submit, then the system blocks submission and highlights unfilled stickers. • Given all 54 stickers are assigned, when the user submits, then the cube state is sent for validation using the same rules as camera-scanned input. • Given the user wants to start over, when they click 'Reset', then all 54 stickers are cleared back to an unset state.	Medium	To Do				
6	FR-06a	The system shall detect and extract all 54 sticker colors from the six submitted face images.	US-06a	Extract Sticker Colors from Scanned Images	user who just finished scanning their cube	the system to automatically extract all 54 sticker colors from my six submitted face images	I don't have to manually identify each sticker color myself	As a user who just finished scanning their cube, I want the system to automatically extract all 54 sticker colors from my six submitted face images, so that I don't have to manually identify each sticker color myself.	• Given six submitted face images, when OpenCV processes them, then exactly nine colors per face are extracted and mapped to the 54 sticker positions.	High	In Progress				

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7	FR-06b	The system shall validate that the extracted 54-color cube state is legal and solvable, rejecting invalid configurations with specific feedback.	US-06b	Validate Scanned Cube State Legality	user who just finished scanning their cube	the system to check that my scanned colors form a legal, solvable cube	I find out immediately if I need to rescan a face instead of getting a failed solve later	As a user who just finished scanning their cube, I want the system to check that my scanned colors form a legal, solvable cube, so that I find out immediately if I need to rescan a face instead of getting a failed solve later.	• Given the extracted 54-color state has an illegal color distribution (e.g. more than 9 stickers of one color), when validation runs, then the system rejects the state and tells the user which face(s) to rescan. • Given the extracted state has a legal color count but an unsolvable permutation or orientation, when validation runs, then the system rejects it with a message explaining the cube appears physically inconsistent.	High	In Progress				
8	FR-06c	The system shall display a 2D preview of the validated cube state for user confirmation before solving begins.	US-06c	Confirm Cube State via 2D Preview	user whose scanned cube state has passed validation	to see a 2D preview of my confirmed cube state	I can visually verify the scan is accurate before proceeding to solve	As a user whose scanned cube state has passed validation, I want to see a 2D preview of my confirmed cube state, so that I can visually verify the scan is accurate before proceeding to solve.	• Given the extracted 54-color state passes all legality checks, when validation completes, then the user is shown a 2D preview for final confirmation before solving begins.	High	In Progress				
9	FR-07	The system shall render the confirmed cube state as a fully interactive, rotatable 3D model in the browser.	US-07	View and Rotate Interactive 3D Cube	user reviewing their scanned cube	to freely rotate and inspect a 3D model that mirrors my physical cube exactly	I can verify the scan is correct and get oriented before I start solving	As a user reviewing their scanned cube, I want to freely rotate and inspect a 3D model that mirrors my physical cube exactly, so that I can verify the scan is correct and get oriented before I start solving.	• Given a validated cube state, when the 3D view loads, then all 26 cubies render with colors matching the confirmed 2D preview. • Given the user drags on the 3D view, when they release, then the camera orbits smoothly around the cube without lag on both desktop and mobile Chromium browsers. • Given the user triggers a face rotation via an on-screen button or swipe or keyboard input, when the animation plays, then the visual state and the underlying logical cube state stay in sync. • Given the 3D view is displayed on a screen as narrow as 360px, when it renders, then all controls remain usable without overlapping elements.	High	In Progress				
10	FR-08a	The system shall compute an optimal solution using the Kociemba two-phase algorithm for Auto-Solve mode.	US-08a	Generate Optimal Solving Algorithm (Auto-Solve)	user who has confirmed their scanned cube state and selected Auto-Solve	the system to calculate an optimal solving sequence using the Kociemba two-phase algorithm	I can watch my exact cube get solved automatically with minimal moves	As a user who has confirmed their scanned cube state and selected Auto-Solve, I want the system to calculate an optimal solving sequence using the Kociemba two-phase algorithm, so that I can watch my exact cube get solved automatically with minimal moves.	• Given a validated, solvable cube state, when the user requests Auto-Solve, then the Kociemba two-phase algorithm returns a move sequence within 2 seconds. • Given the solver receives an unsolvable or malformed state, when it attempts to compute, then it returns a clear error instead of crashing or hanging. • Given a solution is computed, when it is returned to the frontend, then it includes the total move count for later efficiency comparison.	High	In Progress				
11	FR-08b	The system shall compute a beginner's-method solution for Guided Mode.	US-08b	Generate Guided Solving Algorithm	user who has confirmed their scanned cube state and selected Guided Mode	the system to calculate a beginner's-method move list for my exact cube	I can follow it as a step-by-step guide instead of an optimal solution	As a user who has confirmed their scanned cube state and selected Guided Mode, I want the system to calculate a beginner's-method move list for my exact cube, so that I can follow it as a step-by-step guide instead of an optimal solution.	• Given the user selects Guided Mode, when a solution is requested, then the system returns a beginner's-algorithm-based move list instead of a raw optimal solution. • Given the solver receives an unsolvable or malformed state, when it attempts to compute, then it returns a clear error instead of crashing or hanging. • Given a solution is computed, when it is returned to the frontend, then it includes the total move count for later efficiency comparison.	High	In Progress				
12	FR-09	The system shall support an Interactive Solve mode where the user manipulates the 3D cube themselves.	US-09	Solve the Cube Myself with Live Timer	user practicing their own solving method	to make my own moves on the 3D cube while a timer and move counter run live	I can measure my solving speed and efficiency without any hints	As a user practicing their own solving method, I want to make my own moves on the 3D cube while a timer and move counter run live, so that I can measure my solving speed and efficiency without any hints.	• Given the user enters Interactive Mode, when they make the first move, then the timer starts automatically and the move counter increments with every subsequent move. • Given the user reaches a fully solved state, when the last move is made, then the timer stops immediately and the final time and move count are recorded. • Given a solve is completed, when results are displayed, then the optimal move count from FR-08a is shown alongside the user's move count for comparison. • Given the user wants to abandon the attempt, when they click 'Reset Cube', then the timer, move counter, and cube state all reset to the original scrambled state.	Medium	To Do				

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13	FR-10	The system shall support a Guided Solve mode that presents move-by-move instructions following traditional beginner's algorithm.	US-10	Follow Step-by-Step Guided Instructions	user learning to solve the cube for the first time	to see one move at a time in a sidebar and apply it on my physical cube before advancing	I can learn beginner's algorithms without switching attention between a tutorial video and my cube	As a user learning to solve the cube for the first time, I want to see one move at a time in a sidebar and apply it on my physical cube before advancing, so that I can learn beginner's algorithms without switching attention between a tutorial video and my cube.	• Given Guided Mode is active, when the beginner's-algorithm move list loads, then the current move is visually highlighted in the sidebar and mirrored on the 3D model. • Given the user clicks 'Next', when the current move is applied, then the 3D cube animates that single move and advances the highlight to the following move. • Given the user clicks 'Previous', when triggered, then the last applied move is undone on the 3D model and the highlight moves back one step. • Given the user reaches the final move in the list, when it is applied, then the system confirms the cube is fully solved and offers to return to the Main Menu.	Medium	To Do				
14	FR-11	The system shall support an Auto-Solve mode that animates the full computed solution sequence without user input.	US-11	Watch the Cube Solve Itself Automatically	user who wants to see the solution demonstrated	to trigger Auto-Solve and watch the entire move sequence animate on its own	I can study the solving technique passively before attempting it myself	As a user who wants to see the solution demonstrated, I want to trigger Auto-Solve and watch the entire move sequence animate on its own, so that I can study the solving technique passively before attempting it myself.	• Given the user selects Auto-Solve, when they click 'Play', then the full move sequence animates automatically at a consistent, smooth frame rate. • Given the animation is playing, when the user clicks 'Pause', then the animation halts on the current move and can be resumed from the same point. • Given the user wants to review a specific moment, when they drag the progress scrubber, then the 3D cube jumps to the corresponding move in the sequence. • Given the animation reaches the last move, when it completes, then the cube visually matches a fully solved state and playback controls reset. • Given the user is watching an auto solve, when they adjust the speed (slider/button), then the animation speed changes accordingly without requiring them to click next.	Medium	To Do				
15	FR-12a	The system shall track solve time and move count per session in Interactive Mode.	US-12a	Track Solve Time and Move Count	user practicing in Interactive Mode	my solve time and move count to be tracked automatically during my attempt	I have accurate raw data recorded for every attempt without manual timing	As a user practicing in Interactive Mode, I want my solve time and move count to be tracked automatically during my attempt, so that I have accurate raw data recorded for every attempt without manual timing.	• Given a completed solve, when results are calculated, then the recorded time and move count are stored for that session. • Given results are shown, when the user clicks 'Save', then the session record is persisted to their solve history.	Low	To Do				
16	FR-12b	The system shall calculate an efficiency percentage comparing the user's move count to the system-generated solution.	US-12b	View Solve Efficiency Percentage	user who just completed a solve in Interactive Mode	to see my efficiency percentage compared to the system-generated solution	I know exactly how much room for improvement I have	As a user who just completed a solve in Interactive Mode, I want to see my efficiency percentage compared to the system-generated solution, so that I know exactly how much room for improvement I have.	• Given a completed solve, when results are calculated, then efficiency is computed as system-generated move count divided by the user's move count, expressed as a percentage. • Given the calculated efficiency, when it is displayed, then it is shown alongside raw solve time and move count in the results panel. • Given the user's move count is lower than or equal to the system-generated count, when efficiency is computed, then it is capped at 100% rather than exceeding it.	Low	To Do				
17	FR-13a	The system shall persist user solve history server-side.	US-13a	Persist Solve History	user tracking long-term progress	every completed solve to be saved to my account on the server	my history is available no matter which device I log in from	As a user tracking long-term progress, I want every completed solve to be saved to my account on the server, so that my history is available no matter which device I log in from.	• Given a user has completed at least one saved solve, when they open the History page, then every past solve is listed with date, time, move count, and efficiency. • Given a user has solved on both desktop and mobile, when they log in from either device, then their complete history appears identically since it is stored server-side. • Given a user has no completed solves yet, when they open the History page, then the system shows an empty-state message instead of an error.	Low	To Do				
18	FR-13b	The system shall identify and display the user's personal best solve from their history.	US-13b	View Personal Best	user tracking long-term progress	my fastest recorded solve to be flagged as my Personal Best	I can quickly see the benchmark I'm trying to beat	As a user tracking long-term progress, I want my fastest recorded solve to be flagged as my Personal Best, so that I can quickly see the benchmark I'm trying to beat.	• Given a user's history contains multiple solves, when the page loads, then the single fastest solve is flagged as the user's Personal Best.	Low	To Do				

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19	FR-14	The system shall display a global leaderboard ranking users by solve performance.	US-14	Compare My Rank on the Global Leaderboard	competitive user comparing progress against others	to view a global leaderboard ranked by solve time and efficiency, with my own rank highlighted	I stay motivated to practice and can see how I measure up against other users	As a competitive user comparing progress against others, I want to view a global leaderboard ranked by solve time and efficiency, with my own rank highlighted, so that I stay motivated to practice and can see how I measure up against other users.	• Given multiple users have saved solves, when the leaderboard loads, then entries are sorted by best solve time by default, with ties broken by higher efficiency. • Given the logged-in user appears anywhere on the leaderboard, when it renders, then their own row is visually highlighted even if they must scroll to see it. • Given a new personal best is saved, when the leaderboard is next viewed, then it reflects the updated ranking without requiring more than one manual page refresh. • Given the leaderboard has more entries than fit on one screen, when the user scrolls or paginates, then additional ranked entries load correctly in order.	Low	To Do
20	FR-15	The system shall allow a registered user to reset a forgotten password via an emailed reset link.	US-15	Reset Forgotten Password	registered user who has forgotten their password	to request a password reset link sent to my registered email	I can regain access to my account without contacting support	As a registered user who has forgotten their password, I want to request a password reset link sent to my registered email, so that I can regain access to my account without contacting support.	• Given a user enters a registered email on the 'Forgot Password' screen, when they submit it, then a time-limited reset link is emailed to that address. • Given a user enters an email not associated with any account, when they submit it, then the system shows a generic confirmation message without revealing whether the account exists. • Given a user clicks a valid, unexpired reset link, when they submit a new	Medium	To Do
21	FR-16	The system shall generate a randomized, solvable scramble sequence for practice sessions.	US-16	Generate Practice Scramble	user who wants to practice without a physical mixed-up cube	to click 'New Scramble' and have the 3D cube randomize into a fresh, solvable state	I can start a new practice attempt immediately without manually mixing my physical cube	As a user who wants to practice without a physical mixed-up cube, I want to click 'New Scramble' and have the 3D cube randomize into a fresh, solvable state, so that I can start a new practice attempt immediately without manually mixing my physical cube.	• Given the user clicks 'New Scramble', when the request is processed, then the system generates a sequence of at least 20 random face turns and applies it to the 3D cube. • Given a scramble is generated, when it is applied, then the resulting cube state is guaranteed solvable, matching the same legality rules used for camera-scanned states. • Given a new scramble is requested, when it is generated, then the exact	Low	To Do
22	FR-17	The system shall automatically calibrate color detection for each face by reading the center cubelet as the reference color.	US-17	Auto-Calibrate Color via Center Cubelet	user scanning a cube under varying lighting conditions	the system to automatically determine the base color of each face from its center cubelet	I don't have to manually calibrate colors or worry about lighting differences across faces	As a user scanning a cube under varying lighting conditions, I want the system to automatically determine the base color of each face from its center cubelet, so that I don't have to manually calibrate colors or worry about lighting differences across faces.	• Given a user scans a face, when the image is processed, then the system identifies the color of the center cubelet and uses it as the reference for that face's color. • Given the center cubelet color is determined, when the other 8 cubelets are evaluated, then their colors are classified relative to the 6 reference center colors. • Given the system relies on center cubelets, when no manual calibration	High	In Progress
23	FR-18	The system shall allow a user to undo and redo individual moves during an Interactive Mode solve.	US-18	Undo and Redo Moves During Interactive Solve	user practicing their own solving method in Interactive Mode	to undo my last move or redo a move I just undid	I can correct an accidental turn without resetting my entire attempt	As a user practicing their own solving method in Interactive Mode, I want to undo my last move or redo a move I just undid, so that I can correct an accidental turn without resetting my entire attempt.	• Given at least one move has been made in the current attempt, when the user clicks 'Undo', then the 3D cube reverts to the state before that move and the move counter decrements by one. • Given a move has just been undone, when the user clicks 'Redo', then the undone move is reapplied and the move counter increments back. • Given the user makes a brand-new move after undoing one or more	Low	To Do
24	FR-19	The system shall let a user export or share a summary of a completed solve, including time, move count, and efficiency.	US-19	Share Solve Result	user who just completed a solve they are proud of	to generate a shareable summary card of my time, move count, and efficiency	I can post or send my result outside the platform without manually re-typing the numbers	As a user who just completed a solve they are proud of, I want to generate a shareable summary card of my time, move count, and efficiency, so that I can post or send my result outside the platform without manually re-typing the numbers.	• Given a completed and automatically saved solve, when the user clicks 'Share Result', then the system generates an image or link summarizing time, move count, efficiency, and solve mode. • Given a share link is generated, when it is opened by anyone, then it displays a read-only summary without exposing the viewer's own account data. • Given the user clicks 'Copy Link' or 'Download Image', when the action	Low	To Do
25	FR-20	The system shall allow a user to perform cube face rotations using keyboard shortcuts, mapping standard cube notation keys (U, D, L, R, F, B) to clockwise turns and their Shift-modified versions to counter-clockwise turns.	US-20	Control Cube with Keyboard Shortcuts	user practicing their own solving method in Interactive Mode	to press notation keys like R, U, L, D, F, B (and Shift+key for the reverse direction) to rotate cube faces instead of clicking	I can turn the cube as fast as I would with a physical speedcube, without breaking my flow to reach for a mouse	As a user practicing their own solving method in Interactive Mode, I want to press notation keys like R, U, L, D, F, B (and Shift+key for the reverse direction) to rotate cube faces instead of clicking, so that I can turn the cube as fast as I would with a physical speedcube, without breaking my flow to reach for a mouse.	• Given Interactive Mode is active and the cube has focus, when the user presses R, then the right face rotates 90° clockwise and the move is logged in the move counter as "R". • Given Interactive Mode is active, when the user presses Shift+R, then the right face rotates 90° counter-clockwise and the move is logged as "R'". • Given the user presses any of U, D, L, R, F, B (with or without Shift), when the	Medium	To Do
26	FR-21a	The system shall detect when a user crosses a predefined milestone threshold (cumulative solve count or individual solve time) and award the corresponding badge.	US-21a	Earn Milestone Achievement Badge	returning user tracking long-term progress	to automatically receive a badge when I hit a milestone, such as completing my 10th solve or finishing a solve in under 100 seconds	I stay motivated by visible markers of progress instead of only comparing raw numbers	As a returning user tracking long-term progress, I want to automatically receive a badge when I hit a milestone, such as completing my 10th solve or finishing a solve in under 100 seconds, so that I stay motivated by visible markers of progress instead of only comparing raw numbers.	• Given a user completes and saves their 10th total solve, when the save is processed, then the "10 Solves" badge is awarded to their account and a congratulatory notification is shown. • Given a user completes and saves a solve with a recorded time under 100 seconds, when the save is processed, then the "Sub-100" badge is awarded if the user does not already have it. • Given a user has already earned a specific badge, when they complete	Low	To Do

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27	FR-21b	The system shall display all badges earned by a user on their profile, ordered by date unlocked.	US-21b	View Earned Badges on Profile	user who has earned one or more badges	to see all my earned badges on my profile, in the order I unlocked them	I can visibly track my progression over time	As a user who has earned one or more badges, I want to see all my earned badges on my profile, in the order I unlocked them, so that I can visibly track my progression over time.	• Given a user views their profile, when the page loads, then all earned badges are displayed in the order they were unlocked, each with the date earned.	Low	To Do				
28	FR-21c	The system shall retroactively evaluate existing users' historical solves when an admin adds a new milestone threshold.	US-21c	Retroactive Badge Evaluation on New Threshold	admin adding a new milestone threshold	existing users' historical solves to be automatically re-evaluated against the new threshold	already-qualifying users receive the new badge without needing to complete a new solve	As an admin adding a new milestone threshold, I want existing users' historical solves to be automatically re-evaluated against the new threshold, so that already-qualifying users receive the new badge without needing to complete a new solve.	• Given the system defines milestone thresholds (e.g. 10, 50, 100, 500 solves; sub-100s, sub-60s, sub-30s times), when a new threshold is added by an admin, then existing users' historical solves are retroactively evaluated so already-qualifying users receive the new badge without needing a new solve.	Low	To Do				
29	FR-22	The system shall allow a user to log out of their account from any page.	US-22	Log Out of Account	logged-in user	to log out of my account	I can end my session and protect my account on shared devices	As a logged-in user, I want to log out of my account, so that I can end my session and protect my account on shared devices.	• Given a logged-in user clicks 'Log Out', when the action is confirmed, then their session is terminated and they are redirected to the login screen. • Given a user has logged out, when they try to access a protected page using the browser back button, then they are redirected to the login screen instead of seeing cached content. • Given a user is mid-solve in Interactive Mode when they log out, when they log back in, then their in-	Low	To Do				